Final mini project, SAP-1

Grouping

A group of 5 students.

What SAP-1 is

Simple-as-possible-1, or SAP-1, is an extremely simplified computer architecture designed for educational purposes only.

Specifications

- An 8-bit, single bus, and hardwired system.
- The overall memory size is 16 bytes.
- An arithmetic logic unit, or ALU, is included.
- The included registers are as follows.
    - Program counter, or PC.
    - Memory address register, or MAR.
    - Instruction register, or IR.
    - The A register to store an operation operand and the result.
    - The B register to store another operation operand.

Memory assumption

Address	Data
3	4
6	5

Instructions

1. Implement SAP-1 using Verilog.
2. Write a report about what happens when you run the assembly instructions below on your implementation.
    a. LDA 3
    b. ADD 6

3. Include a screenshot, or screenshots, of the waveform from the Vivado simulation.

Module breakdown

- Top module, CPU.
- Control unit.
- Register.
- Program counter, PC.
- Arithmetic logic unit, ALU.
- Memory.
- Bus.

Note

- In the report, explain in timing steps. Additionally, don't just say what happens in each timing step. Instead, reference corresponding control signals, and what happens when each one changes its value to something. This is the **most important** note.
- The given assembly instructions are just for you to have a reference when explaining, and for the grader to know whether or not you really understand your code. However, the project will not be graded based on them alone, so make sure your implementation works for random tests as well.
- The ARM equivalent of the assembly instructions above is as follows.
    a. LDR A, [3]
    b. ADD A, [6]
- You're allowed to break your module down differently. If you do, include what each of your modules does in the report.

Submission in LEB2

1. Your report in PDF.
2. A zip file containing all of your source codes and testbenches.